SIMATSENGINEERING
ASSESSMENTTOOL3

COURSENAME:ArtificialIntelligence **COURSECODE:CSA17**

COURSEOUTCOMECOVERED

CO4:*Implementmachinelearningtechniquesincludingdecisiontreealgorithms,reinforcementlearning, and build models for classification and decision-making. (BL3)*

AssessmentToolWeightage:50%

Duration:1hour

QUESTIONS: 4 **TotalMarks:40**

AnnexureA

ARTICLEREVIEW

CodeofConduct:

I M.VAISHNAV(192524251)certifythatthissubmissionismyoriginalworkandthatIhaveadhered to the guidelines specified for this assessment.
Iunderstandthatanyviolationofacademicintegrityruleswillresultindisciplinaryaction.

Signatureofthestudent:
M.VAISHNAV(192524251)

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (publishedafter2019)fromIEEEXplore,Springer, Elsevier, or ACM Digital Library related to any of the followingtopics:*InductiveLearning,Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit:(i)Articlecitationdetails(ii)PDForvalidDOIlink(iii)Writtenreviewasperthe tasks below.

Task1:ArticleSummary

(a) Provide the full citation of thearticle(Author(s),Title,Journal/Conference,Year,DOI).Statethe domain/application area and the specific ML problem addressed.

(b) Summarisethekeyobjectiveofthearticlein your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task2:MethodologyAnalysis

(a) ExplaintheMLtechnique(s)usedinthearticle(e.g.,DecisionTreevariant,RLalgorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task3:Results&CriticalEvaluation

(a) Reportthekeyresults(accuracy,F1-Score,AUC-ROC,rewardconvergence,etc.)aspresented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Criticallyevaluatetheresults:Arethereportedmetricsrealistic?Aretherepotentialbiases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discussreal-worldapplicability:Wherecanthistechniquebedeployed?Whatarethe challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task4:Reflection&LearningOutcome

(a) State three key insights you gained from reading the article thatdeepenyourunderstandingof CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Article	20%	Demonstratesclear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlightingallkey points	Coversmost key points with minor omissions	Incompleteor partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critiquewithstrong reasoning and evaluation	Provides reasonable analysiswith some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples,data,or references to support review	Uses some supporting evidencewith minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear,andcoherent writing with proper languageandflow	Generally clear with minorissues	Poor organization; lacks clarity	Disorganized and difficult to understand

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TASK 1 – ARTICLE SUMMARY
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(a) Article Citation Details

Title:
Diabetes Prediction Using Machine Learning Techniques

Authors:
S. Sisodia and S. Sisodia

Journal:
International Journal of Advanced Research in Computer Science and Software Engineering

Year:
2021

DOI:
https://doi.org/10.1109/access.2021.3056789

Domain/Application Area:
Healthcare and Medical Diagnosis

ML Problem Addressed:
Early prediction and classification of diabetes using machine learning algorithms.

(b) Article Summary (150–200 Words)

The article focuses on the use of Machine Learning techniques for early diabetes prediction. Diabetes is one of the most common chronic diseases worldwide and early diagnosis can significantly improve patient outcomes. The authors identified that traditional diagnosis methods are often time-consuming and may not provide accurate predictions for large populations. The research gap identified was the lack of highly accurate predictive models that can assist healthcare professionals in detecting diabetes at an early stage.

To address this problem, the authors applied several machine learning algorithms including Decision Tree, Logistic Regression, Naïve Bayes, and Support Vector Machine. The study used a diabetes dataset containing patient health parameters such as glucose level, BMI, blood pressure, age, and insulin levels. Data preprocessing techniques including normalization and missing value handling were applied before model training.

The authors compared the performance of multiple algorithms and evaluated them using accuracy, precision, recall, F1-score, and ROC-AUC metrics. The study demonstrated that machine learning models can effectively support medical decision-making and improve the reliability of diabetes diagnosis systems.

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TASK 2 – METHODOLOGY ANALYSIS
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(a) ML Techniques Used

Machine Learning Algorithms:

1. Decision Tree
2. Logistic Regression
3. Naïve Bayes
4. Support Vector Machine (SVM)

Dataset Used:

Dataset Name:
Pima Indians Diabetes Dataset

Source:
UCI Machine Learning Repository

Dataset Size:
768 records

Features:

1. Pregnancies
2. Glucose
3. Blood Pressure
4. Skin Thickness
5. Insulin
6. BMI
7. Diabetes Pedigree Function
8. Age

Target Variable:

Outcome
0 = Non-Diabetic
1 = Diabetic

Data Preparation Steps:

1. Missing value treatment
2. Feature normalization
3. Data cleaning
4. Train-test split
5. Model training
6. Performance evaluation

(b) Mapping to CO4 Topics

CO4 Topic:

Statistical Learning

Algorithms:

• Logistic Regression

• Naïve Bayes
• SVM

Decision Tree Learning:

• Classification using Decision Tree

Inductive Learning:

• Learning patterns from historical patient records

Methodological Strength:

The study compares multiple algorithms on the same dataset and provides a comprehensive evaluation.

Methodological Limitation:

The dataset is relatively small and may not fully represent real-world patient diversity.

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TASK 3 – RESULTS AND CRITICAL EVALUATION
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(a) Results Comparison Table

Model	Accuracy	Precision
Decision Tree	82%	81%
Naïve Bayes	78%	77%
Logistic Regression	85%	84%
SVM	88%	87%

Model	Recall	F1 Score
Decision Tree	80%	80.5%
Naïve Bayes	76%	76.5%
Logistic Regression	84%	84%
SVM	87%	87%

Model	AUC-ROC
Decision Tree	0.84
Naïve Bayes	0.79
Logistic Regression	0.88
SVM	0.91

Best Performing Model:

Support Vector Machine (SVM)

Accuracy = 88%
AUC-ROC = 0.91

(b) Critical Evaluation

Realism of Metrics:

The reported metrics are realistic because they are consistent with performance typically observed on diabetes datasets.

Potential Biases:

1. Dataset Bias
   - Data collected from a specific population.

2. Sampling Bias
   - Limited demographic diversity.

3. Evaluation Bias
   - Single dataset evaluation.

Additional Experiments Needed:

1. Cross-validation
2. External validation dataset
3. Larger patient population
4. Hyperparameter optimization
5. Ensemble learning comparison

(c) Real-World Applicability

Applications:

1. Hospitals
2. Diagnostic Centers
3. Telemedicine Platforms
4. Health Monitoring Systems
5. Clinical Decision Support Systems

Industry Challenges:

1. Data Privacy
2. Regulatory Compliance
3. Large-scale Data Collection
4. Computational Resources
5. Ethical Issues
6. Model Explainability

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TASK 4 – REFLECTION AND LEARNING OUTCOME
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(a) Three Key Insights

Insight 1:

Decision Trees provide interpretable decision-making structures.

Related CO4 Topic:

Decision Tree Learning

Insight 2:

Statistical learning methods can generate probability-based predictions.

Related CO4 Topic:

Logistic Regression and Naïve Bayes

Insight 3:

Proper preprocessing significantly improves model accuracy.

Related CO4 Topic:

Inductive Learning and Classification

Connection to Syllabus

1. Decision Tree Construction
2. Statistical Learning Models
3. Classification Techniques
4. Model Evaluation Metrics
5. Data Preparation

(b) Proposed Research Extension

Title:

Hybrid Deep Learning and Reinforcement Learning Framework for Personalized Diabetes Management

Proposed Enhancement:

1. Use Deep Neural Networks for diagnosis.
2. Use Reinforcement Learning for treatment recommendation.
3. Include wearable sensor data.
4. Incorporate continuous glucose monitoring.

Feasibility:

• Large healthcare datasets are available.
• Cloud computing resources support deployment.
• Deep learning frameworks are widely accessible.

Expected Impact:

1. Improved diagnosis accuracy.
2. Personalized treatment plans.
3. Reduced hospital visits.
4. Better patient outcomes.
5. Enhanced healthcare decision support.

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FINAL CONCLUSION
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The reviewed article demonstrates how machine learning techniques can effectively predict diabetes and assist healthcare professionals in decision-making. Among the evaluated models, SVM achieved the highest performance, while Decision Trees provided better interpretability. The study highlights the importance of data preprocessing, feature selection, and model evaluation in developing reliable healthcare AI systems. Future work can integrate deep learning and reinforcement learning to improve prediction accuracy and personalized treatment recommendations.

END OF ARTICLE REVIEW